

Python Programming

Unit 03 – Lecture 04: Regular Expressions (Regex) in Python

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

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- Useful for:
 - validation (email, phone, ID formats)
 - extraction (find all numbers, dates, names)
 - cleaning (replace multiple spaces, remove symbols)

Important: Raw Strings for Patterns

- Python uses backslashes in strings (escape sequences)

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Important: Raw Strings for Patterns

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- Regex also uses backslashes (e.g., `\d`)
- Use raw strings: `r"..."`

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- Anchors: `^` start, `$` end

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- `re.findall(p, text)`: all matches as a list
- `re.sub(p, repl, text)`: replace matches

Example: Validate a Phone Number (10 digits)

```
import re
phone = input("Phone: ").strip()
if re.fullmatch(r"\d{10}", phone):
    print("Valid")
else:
    print("Invalid")
```

Example: Extract Emails

```
import re
text = "Mail us at a@b.com or help@upes.ac.in"
emails = re.findall(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+")
print(emails)
```


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 - email addresses
 - Indian-style 10-digit phone numbers
- Demonstrates: `findall`, `fullmatch`, `sub`

Checkpoint 1

Question: What is the difference between `search` and `fullmatch`?

Checkpoint 2

Question: Write a regex to match a 4-digit PIN code (e.g., 248001). Should it allow leading zeros?

Think-Pair-Share

Discuss:

- When is regex a good tool?
- When is plain string logic better than regex?

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- Prefer raw strings `r"..."` for patterns
- Learn meta characters and quantifiers (`.`, `[]`, `+`, `{m,n}`)
- Use `fullmatch` for validation and `findall` for extraction

Exit Question

Write a one-line regex (pattern only) for a date in format YYYY-MM-DD.